**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# LESSON 309 · Paper 2 examiner-style feedback

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. Why must recursive calls progress?

Q2. What does a midpoint calculate?

Q3. What happens when low>high?

Q4. C# [2,4,6] with a[1]=a[0]+a[2]. Final array?

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# I Do · Read new knowledge

## Key vocabulary

mark scheme — Criteria for awarding marks.

termination — Algorithm eventually stops.

auxiliary space — Additional storage beyond the input.

## New learning

Understand — Identify whether the error lies in problem reading, algorithm state or pseudocode.

Apply — Explain every bound update, recursive return or data-structure operation.

Evaluate — Retest the method with a different input to demonstrate transfer.

## Worked example

Repair a SumTo recursion missing a base case.

1. Explain the result — The n==0 base case stops recursion and returns zero.

# We Do · Practise with guidance

Repair a binary search that can repeat the midpoint forever.

```
if(a[mid]<target)low=mid+1;
else if(a[mid]>target)high=mid-1;
```

Explain the result

The boundaries exclude a failed midpoint, so the remaining interval shrinks.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Question 1

An algorithm's recursion has no base case. Explain fault and give two repairs. [6]

### Question 2

Trace binary search for 9 in [1,3,5,7,9], recording every low, high and midpoint. [6]

### Question 3

Correct: 'merge sort has O(1) auxiliary space because the temp array is short-lived'. [6]

# Exit Ticket · Answer before checking

Getting a correct output once guarantees an algorithm is bug-free.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

### Page 1 onwards · Do Now

Q1. To reach the base case.

Q2. The middle index of remaining interval.

Q3. No candidates remain; not found.

Q4. [2,8,6].

### You Do pages · Independent questions

Q1. Unbounded calls [2], possible stack overflow [1]; correct base [2], parameter changes towards base [1].

Q2. low0 high4 mid2→5 [2], low3 high4 mid3→7 [1], low4 high4 mid4→9 [2], returns index4 [1].

Q3. Typical array merge needs O(n) peak auxiliary storage [3]; worst-case time O(n log n) [2]; temporary lifetime does not change peak Big O memory [1].

### Exit Ticket · Model answer

False. Test edge cases and varied inputs.

# Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**